by the target molecule. Inelastic scattering of impinging photons where some energy is lost to (or gained from) the target molecule returns scattered light with a higher (or lower) wavelength depending upon whether energy was lost to (or gained from) the target molecule.

The difference is dictated by the energy of vibrational modes of the target molecule and therefore constitutes the fingerprint or basis of identification. Complex mixtures are identified using algorithms for pattern recognition. Fig. 11 shows a typical Raman spectroscopy setup.

P.Eye (Fig. 12) developed by Swedish company Portendo in agreement with BAE is a handheld Raman explosives detector that can be used for the detection of roadside bombs and IEDs.

ObserveR of SciAps Inc. (Fig. 13) is another portable Raman system designed for stand-off detection and identification of explosives, toxic industrial chemicals (TICs) and toxic industrial materials (TIMs) from distances ranging from 0.1m to 3m. It has an onboard library of Raman signatures of a large number of explosive samples, TICs and TIMs, and a provision for new user-programmable entries.

Inspector-300 (Fig. 14) and Inspector-500 are other portable Raman detectors, respectively, operating at 785nm and 1030nm, from the same company. These are particularly suited for the detection of samples that produce high levels of fluorescence.

ReporteR (Fig. 15) is a handheld Raman system from SciAps Inc. It is a miniaturised version of Inspector-300 system.

Advantage series bench-top Raman systems from SciAps Inc. offers four variants operating at 532nm, 633nm, 785nm and 1064nm.

THOR-1064 (Fig. 16) of Chemring Detection Systems is another Raman-based handheld explosive detection system. It employs a 500mW laser operating at 1064nm. It illuminates an unknown sample with laser radiation. Return signal is digitised, and Raman signature unique to the sample is recorded. An onboard processor compares the sample's Raman signature to a stored library of Raman spectra of known compounds. A correlation algorithm identifies the best match and displays it on the operator's console. Operation at 1064nm offers considerable fluorescence reduction over 785nm wavelength systems.

Laser Science and Technology Centre (LASTEC) of Defence Research and Development Organization (DRDO) has developed a portable Raman-based explosive detector. The system named Preemptor (Fig. 17) operates at 532nm. It can detect and identify explosive samples in solid and liquid forms from stand-off distances of up to 5m.

The system has potential applications in screening of liquid explosives in transparent bottles and explosive threats in the form of powder and pills. The system has been evaluated against a large number of common explosive samples, including DNT, TNT, TATP, RDX and HMX. Reportedly, the system has undergone user evaluation with paramilitary forces.

LIF is another important tool for similar applications. Though a valuable tool in combustion diagnostics and studying decomposition of explosives, it has not been found useful for the detection of explosive agents.

Almost all laser-based stand-off detection methods have detection limits too high to make these suitable for the detection of explosives in field conditions. Such detection requires the capability to detect traces in vapour phase or in the form of particles.

Another problem is insufficient selectivity to identify target explosives in the presence of interferents. In fact, none of the laser-based spectroscopic methodologies available today are ready for full-functionality prototype manufacturing.

There is lot of scope for further research to improve the performance of more mature technologies in terms of both detection sensitivity and selectivity. The final solution would probably lie not in one type of sensor, but in the integration of several technologies to derive the benefits of the best of each. Simultaneous use of stand-off LIBS and stand-off Raman has been exploited successfully.

To be concluded